

## ProgDS-2025-10-14 Exercise – 6: Classification

In this Exercise you will be processing the following datasets:

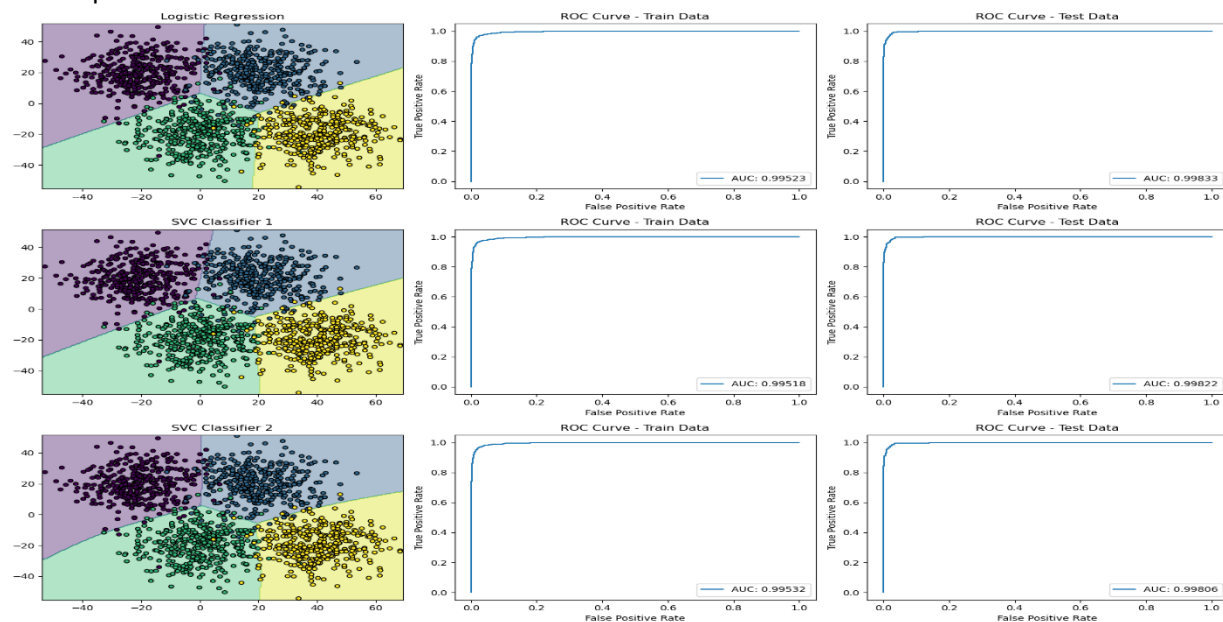
- Clusters-4-v0.csv
- Clusters-4-v1.csv
- Clusters-4-v2.csv

### Part-1:

1. Divide each data set into ‘train’ and ‘test’ datasets, once, and use them for all subsequent steps.
2. Review the data using appropriate plots and understand the overall structure of the data. Comment on the data and anticipate how well LogisticRegression will perform on the data.
3. Use the following algorithms / variants to process the datasets:
  - a. Logistic Regression
  - b. Logistic Regression after adding polynomial features
  - c. SVC – with ‘linear’ kernel (what is ‘linear’?)
  - d. SVC – with ‘rbf’ kernel (what is ‘rbf’?)
  - e. Random Forest Classifier – for various combinations of “minimum observations per leaf” (say 1 to 5) and “tree depth” (say 2 to 5). This of this as a “grid search” problem, where a model is created for every combination of these parameters.
  - f. Neural Network Classifier – with hidden\_layer\_sizes=(5)
  - g. Neural Network Classifier – with hidden\_layer\_sizes=(5,5)
  - h. Neural Network Classifier – with hidden\_layer\_sizes=(5,5,5)
  - i. Neural Network Classifier – with hidden\_layer\_sizes=(10)
4. In each of the above cases generate, capture, and save all the results, for all the datasets, into a common csv file – to facilitate analysis later on. The following metrics (for train and test data) should be created: (For example, see the image that follows):
  - Accuracy, Precision (per class), Precision (average), Recall (per class), Recall (average), F1-score (per class), F1-score (average), AUC (per class), AUC (average).
  - (Hint: The following functions may be used: accuracy\_score, precision\_score, recall\_score, f1\_score, roc\_auc\_score, roc\_curve)

| algorithm_name              | train_or_test_data | accuracy | precision_1 | precision_2 | precision_3 | precision_4 | precision_avg | recall_1 | recall_2 | recall_3 | recall_4 | recall_avg | F1_1   | F1_2   | F1_3   | F1_4   | F1_avg | AUC_1  | AUC_2  | AUC_3  | AUC_4  | AUC_avg |
|-----------------------------|--------------------|----------|-------------|-------------|-------------|-------------|---------------|----------|----------|----------|----------|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| Logistic Regression         | train              | 0.9514   | 0.9521      | 0.9470      | 0.9373      | 0.9690      | 0.9513        | 0.9521   | 0.9404   | 0.9406   | 0.9723   | 0.9513     | 0.9521 | 0.9437 | 0.9389 | 0.9706 | 0.9513 | 0.9960 | 0.9955 | 0.9923 | 0.9972 | 0.9952  |
| Logistic Regression         | test               | 0.9549   | 0.9710      | 0.9722      | 0.9333      | 0.9444      | 0.9553        | 0.9853   | 0.9333   | 0.9459   | 0.9577   | 0.9556     | 0.9781 | 0.9524 | 0.9396 | 0.9510 | 0.9553 | 0.9996 | 0.9976 | 0.9978 | 0.9984 | 0.9983  |
| SVC Classifier 1            | train              | 0.9540   | 0.9556      | 0.9505      | 0.9406      | 0.9690      | 0.9539        | 0.9589   | 0.9439   | 0.9406   | 0.9723   | 0.9539     | 0.9573 | 0.9472 | 0.9406 | 0.9706 | 0.9539 | 0.9956 | 0.9954 | 0.9924 | 0.9973 | 0.9952  |
| SVC Classifier 1            | test               | 0.9549   | 0.9710      | 0.9722      | 0.9333      | 0.9444      | 0.9553        | 0.9853   | 0.9333   | 0.9459   | 0.9577   | 0.9556     | 0.9781 | 0.9524 | 0.9396 | 0.9510 | 0.9553 | 0.9995 | 0.9976 | 0.9974 | 0.9983 | 0.9982  |
| SVC Classifier 2            | train              | 0.9497   | 0.9583      | 0.9443      | 0.9247      | 0.9719      | 0.9498        | 0.9452   | 0.9509   | 0.9441   | 0.9585   | 0.9497     | 0.9517 | 0.9476 | 0.9343 | 0.9652 | 0.9497 | 0.9967 | 0.9955 | 0.9920 | 0.9970 | 0.9953  |
| SVC Classifier 2            | test               | 0.9583   | 0.9571      | 0.9595      | 0.9351      | 0.9851      | 0.9592        | 0.9853   | 0.9467   | 0.9730   | 0.9296   | 0.9586     | 0.9710 | 0.9530 | 0.9536 | 0.9565 | 0.9585 | 0.9995 | 0.9969 | 0.9975 | 0.9982 | 0.9981  |
| Random Forest Classifier 1  | train              | 1.0000   | 1.0000      | 1.0000      | 1.0000      | 1.0000      | 1.0000        | 1.0000   | 1.0000   | 1.0000   | 1.0000   | 1.0000     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000  |
| Random Forest Classifier 1  | test               | 0.9410   | 0.9054      | 0.9571      | 0.9583      | 0.9444      | 0.9413        | 0.9853   | 0.8933   | 0.9324   | 0.9577   | 0.9422     | 0.9437 | 0.9241 | 0.9452 | 0.9510 | 0.9410 | 0.9989 | 0.9903 | 0.9939 | 0.9982 | 0.9953  |
| Random Forest Classifier 2  | train              | 0.9627   | 0.9723      | 0.9514      | 0.9512      | 0.9757      | 0.9627        | 0.9623   | 0.9614   | 0.9545   | 0.9723   | 0.9626     | 0.9673 | 0.9564 | 0.9529 | 0.9740 | 0.9626 | 0.9993 | 0.9987 | 0.9983 | 0.9996 | 0.9990  |
| Random Forest Classifier 2  | test               | 0.9479   | 0.9437      | 0.9577      | 0.9467      | 0.9437      | 0.9479        | 0.9853   | 0.9067   | 0.9595   | 0.9437   | 0.9488     | 0.9640 | 0.9315 | 0.9530 | 0.9437 | 0.9481 | 0.9989 | 0.9926 | 0.9969 | 0.9984 | 0.9967  |
| Random Forest Classifier 3  | train              | 0.9583   | 0.9655      | 0.9507      | 0.9446      | 0.9723      | 0.9583        | 0.9589   | 0.9474   | 0.9545   | 0.9723   | 0.9583     | 0.9622 | 0.9490 | 0.9496 | 0.9723 | 0.9583 | 0.9988 | 0.9980 | 0.9972 | 0.9991 | 0.9983  |
| Random Forest Classifier 3  | test               | 0.9583   | 0.9710      | 0.9595      | 0.9467      | 0.9571      | 0.9586        | 0.9853   | 0.9467   | 0.9595   | 0.9437   | 0.9588     | 0.9781 | 0.9530 | 0.9530 | 0.9504 | 0.9586 | 0.9989 | 0.9936 | 0.9972 | 0.9985 | 0.9970  |
| Neural Network Classifier 1 | train              | 0.9071   | 0.9422      | 0.9485      | 0.9508      | 0.8129      | 0.9136        | 0.9486   | 0.9053   | 0.8112   | 0.9619   | 0.9068     | 0.9454 | 0.9264 | 0.8755 | 0.8811 | 0.9071 | 0.9937 | 0.9915 | 0.9710 | 0.9802 | 0.9841  |
| Neural Network Classifier 1 | test               | 0.9028   | 0.9577      | 0.9710      | 0.9375      | 0.7738      | 0.9100        | 1.0000   | 0.8933   | 0.8108   | 0.9155   | 0.9049     | 0.9784 | 0.9306 | 0.8696 | 0.8387 | 0.9043 | 0.9999 | 0.9942 | 0.9829 | 0.9736 | 0.9876  |
| Neural Network Classifier 2 | train              | 0.9219   | 0.9507      | 0.8856      | 0.9598      | 0.9010      | 0.9243        | 0.9247   | 0.9509   | 0.8357   | 0.9758   | 0.9217     | 0.9375 | 0.9171 | 0.8935 | 0.9369 | 0.9212 | 0.9878 | 0.9833 | 0.9693 | 0.9896 | 0.9825  |
| Neural Network Classifier 2 | test               | 0.9236   | 0.9714      | 0.9452      | 0.9538      | 0.8375      | 0.9270        | 1.0000   | 0.9200   | 0.8378   | 0.9437   | 0.9254     | 0.9855 | 0.9324 | 0.8921 | 0.8874 | 0.9244 | 1.0000 | 0.9965 | 0.9873 | 0.9833 | 0.9918  |
| Neural Network Classifier 3 | train              | 0.9453   | 0.9589      | 0.9462      | 0.9301      | 0.9458      | 0.9452        | 0.9589   | 0.9263   | 0.9301   | 0.9654   | 0.9452     | 0.9589 | 0.9362 | 0.9301 | 0.9555 | 0.9452 | 0.9932 | 0.9916 | 0.9886 | 0.9962 | 0.9924  |
| Neural Network Classifier 3 | test               | 0.9514   | 0.9706      | 0.9722      | 0.9221      | 0.9437      | 0.9521        | 0.9706   | 0.9333   | 0.9595   | 0.9437   | 0.9518     | 0.9706 | 0.9524 | 0.9404 | 0.9437 | 0.9518 | 0.9994 | 0.9971 | 0.9970 | 0.9969 | 0.9976  |

- For each dataset generate plots like the ones below - to understand the classification boundaries and the overall performance of the classifiers:



- Compare the metrics within and across the datasets (train and test) and algorithms. In addition to the variations in metrics, compare aspects related to classification boundaries, overfitting, etc.

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